

ON THE DESIGN OF BFFG

CAMILLA & FRANK (VU AMSTERDAM)

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ABSTRACT. Backward Filtering Forward Guiding is an algorithm for obtaining (weighted) samples of a partially observed stochastic process on a directed acyclic graph. For simplicity, we assume the state-space setting. The backward filtering step yields a sequence of *likelihood potentials* $\{g_i\}$ which are used in forward stochastic simulation of a *guided process* that is informed by these potentials. In this paper we investigate how properties of $\{g_i\}$ affect the efficiency of stochastic simulation of the guided process.

1. INTRODUCTION

We consider a latent discrete-time Markov process $(x_1, \dots, x_n)$. The edge from 0 to 1 fixes a prior on the initial state x_1 . At times $i = 1, \dots, n$ we assume an observation $v_i \mid x_i$ is generated from a distribution with density $k(x_i, v_i)$. We aim to sample from $x_{1:n} \mid v_{1:n}$. Let $h_i(x_i)$ be the density of $v_{i:n} \mid x_i$. The Markov process that is conditioned on $v_{1:n}$ is a Markov process with transition densities

$$p^*(x_i \mid x_{i-1}) = \frac{p(x_i \mid x_{i-1})h_i(x_i)}{\int p(x_i \mid x_{i-1})h_i(x_i) dx_i}.$$

As the sequence of functions $\{h_i\}$ is often not available in closed form, we replace it with a sequence of *guiding* functions $\{g_i\}$. The resulting *guided* process has transition

densities

$$p^\circ(x_i | x_{i-1}) = \frac{p(x_i | x_{i-1})g_i(x_i)}{\int p(x_i | x_{i-1})g_i(x_i) dx_i}$$

The smoothing density is given by

$$\begin{aligned} p^\star(x_{0:n}) &= \prod_{i=0}^n p^\star(x_i | x_{i-1}) = \prod_{i=0}^n \frac{p(x_i | x_{i-1})h_i(x_i)}{\int p(x_i | x_{i-1})h_i(x_i) dx_i} \\ &= \frac{1}{h_0(x_0)} \left[\prod_{i=1}^n k(x_i, v_i) \right] \prod_{i=0}^n p(x_i | x_{i-1}), \end{aligned}$$

where $p(x_0 | x_{-1}) \equiv p(x_0)$. The final equality is shown in Theorem 6 in [VdMSS25]. Under the guided process, we have

$$p^\circ(x_{0:n}) = \prod_{i=0}^n p^\circ(x_i | x_{i-1}) = \prod_{i=0}^n \frac{p(x_i | x_{i-1})g_i(x_i)}{\int p(x_i | x_{i-1})g_i(x_i) dx_i},$$

Therefore

$$\frac{p^\star(x_{0:n})}{p^\circ(x_{0:n})} = \frac{1}{h_0(x_0)} \Psi(x_{0:n}), \quad (1)$$

where

$$\Psi(x_{0:n}) = \left[\prod_{i=1}^n k(x_i, v_i) \right] \prod_{i=0}^n \frac{\int p(x_i | x_{i-1})g_i(x_i) dx_i}{g_i(x_i)}.$$

In particular, the likelihood is given by

$$h_0(x_0) = \mathbb{E}^\circ \Psi(x_{0:n}). \quad (2)$$

1.1. Research questions / goals.

- (1) Bound $\text{KL}(\mathbb{P}^\star || \mathbb{P}^\circ)$ by a discrepancy measure between $\{g_i\}$ and $\{h_i\}$. The behaviour of this KL-divergence characterises the quality of $\mathbb{P}^\circ$ as importance sampling distribution to sample from $\mathbb{P}^\star$ (Cf. [CD18]).
- (2) The right-hand-side of (2) can be approximated by Monte-Carlo simulation from samples of the guided process. Under which conditions is this estimator of finite variance? If we have a candidate $\{g_i\}$ which is not guaranteed to have this property, can we fix this by transforming each g_i ?
- (3) As we can view the guided process as a proposal in a guided Feynman-Kac model, can we use general results from particle filtering to study the quality of the guided process?

2. FORMULATION AS GUIDED FEYNMAN-KAC MODEL

In the notation of Chapter 5 in [CP20], we have a guided Feynman-Kac model with

$$\begin{aligned} \mathbb{M}_0(dx_0) &= p^\circ(x_0) dx_0 \\ M_t(x_{t-1}, dx_t) &= p^\circ(x_t | x_{t-1}) dx_t \\ f_t(v_t | x_t) &= k(x_t, v_t) \\ G_t(x_{t-1}, x_t) &= \frac{k(x_t, v_t)p(x_t | x_{t-1})}{p^\circ(x_t | x_{t-1})} = k(x_t, v_t) \frac{\int p(x_t | x_{t-1})g_t(x_t) dx_t}{g_t(x_t)}. \end{aligned}$$

In a generic particle filtering algorithm, Cf. Algorithm 10.1 in [CP20], the unnormalised weight is given by

$$w_t^n = G_t(X_{t-1}^{A_n}, X_t^n).$$

For particle filtering, by Proposition 11.3 in [CP20], if the potentials G_t are all upper bounded, then there exist constants c'_t such that

$$\mathbb{E} \left[\frac{1}{N} \sum_{n=1}^N W_t^n \varphi(X_t^n) - \mathbb{Q}_t(\varphi) \right]^2 \leq c'_t \frac{\|\varphi\|_\infty^2}{N},$$

where $W_t^n = w_t^n / \sum_{m=1}^N w_t^m$.

The numerator of the expression for $G_t(x_{t-1}, x_t)$ is bounded provided k and g_t are, which is a reasonable assumption in applications. However, the denominator, $g_t(x_t)$, is typically not bounded away from zero. For that reason, we think of *transforming* the maps $\{g_t\}$ to ensure this. For example via

- (A) mapping $g_t(x)$ to $g_t(x) + \epsilon_t$;
- (B) mapping $g_t(x)$ to $\sqrt{g_t(x)^2 + \epsilon_t}$,

where $\{\epsilon_t\}$ is a sequence of strictly positive numbers bounded away from zero. While such transformations on g_t are simple, for BFFG to be efficient, at the same time we need that

- (1) sampling under p° is simple;
- (2) evaluation of $\int p(x_t | x_{t-1}) g_t(x_t) dx_t$ is simple. **I think a positive unbiased estimator would suffice.**

Option (A) ensures (1) as

$$\begin{aligned} p^\circ(x_t | x_{t-1}) &= \frac{p(x_t | x_{t-1})(g_t(x_t) + \epsilon_t)}{\int p(x_t | x_{t-1})(g_t(x_t) + \epsilon_t) dx_t} \\ &= \lambda_t(x_{t-1}) \frac{p(x_t | x_{t-1}) g_t(x_t)}{\int p(x_t | x_{t-1}) g_t(x_t) dx_t} + (1 - \lambda_t(x_{t-1})) p(x_t | x_{t-1}), \end{aligned}$$

where

$$\lambda_t(x_{t-1}) = \frac{\int p(x_t | x_{t-1}) g_t(x_t) dx_t}{\int p(x_t | x_{t-1}) g_t(x_t) dx_t + \epsilon_t}.$$

Hence, the guided process is a mixture distribution of the original guided process and the unconditional process. **use rejection sampling for option (B)?**

3. KL-BOUNDS

Let $\mathbb{P}_T^*$ and $\mathbb{P}_T^\circ$ denote the laws of $X_{0:T}^*$ and $X_{0:T}^\circ$ respectively. According to results in [CD18], the quality of an importance sampling distribution, which is $\mathbb{P}_T^\circ$ here, is dictated by $\text{KL}(\mathbb{P}_T^*, \mathbb{P}_T^\circ)$ given by

$$\text{KL}(\mathbb{P}_T^* \| \mathbb{P}_T^\circ) = \int p^*(x_{0:T}) \log \left(\frac{p^*(x_{0:T})}{p^\circ(x_{0:T})} \right) dx_{0:T}.$$

We would like to bound this by a discrepancy measure between $\{h_i\}$ and $\{g_i\}$.¹ The following lemma is adapted from Proposition 3.1 in [JHJ21].

¹We should keep in mind that the effect of h_i or g_i is invariant under multiplying with a positive constant.

Lemma 1. Define $\xi_t(x) = \log h_t(x) - \log g_t(x)$. Then

$$\log \left(\frac{p^*(x_{0:T})}{p^\circ(x_{0:T})} \right) \leq \sum_{t=0}^T \xi_t(x_t) - \sum_{t=0}^T \int p^\circ(x_t | x_{t-1}) \xi_t(x_t) dx_t$$

Proof. We have

$$\begin{aligned} \log \left(\frac{p^*(x_{0:T})}{p^\circ(x_{0:T})} \right) &= \log \left(\prod_{t=0}^T \frac{p^*(x_t | x_{t-1})}{p^\circ(x_t | x_{t-1})} \right) = \log \left(\prod_{t=0}^T \frac{h_t(x_t) \int p(x_t | x_{t-1}) g_t(x_t) dx_t}{g_t(x_t) \int p(x_t | x_{t-1}) h_t(x_t) dx_t} \right) \\ &= \sum_{t=0}^T \log \left(\frac{h_t(x_t)}{g_t(x_t)} \right) + \sum_{t=0}^T \log \left(\frac{\int p(x_t | x_{t-1}) g_t(x_t) dx_t}{\int p(x_t | x_{t-1}) h_t(x_t) dx_t} \right) \end{aligned}$$

using the convention $p^*(x_0 | x_{-1}) = p(x_0)$ (and analogously for p°). Therefore, each summand in the second summation of satisfies:

$$\begin{aligned} \log \left(\frac{\int p(x_t | x_{t-1}) g_t(x_t) dx_t}{\int p(x_t | x_{t-1}) h_t(x_t) dx_t} \right) &= \frac{\int p(x_t | x_{t-1}) g_t(x_t) dx_t}{\int p(x_t | x_{t-1}) g_t(x_t) dx_t} \cdot \log \left(\frac{\int p(x_t | x_{t-1}) g_t(x_t) dx_t}{\int p(x_t | x_{t-1}) h_t(x_t) dx_t} \right) \\ &\leq \frac{1}{\int p(x_t | x_{t-1}) g_t(x_t) dx_t} \int p(x_t | x_{t-1}) g_t(x_t) \log \left(\frac{p(x_t | x_{t-1}) g_t(x_t)}{p(x_t | x_{t-1}) h_t(x_t)} \right) dx_t \\ &= \int \frac{p(x_t | x_{t-1}) g_t(x_t)}{\int p(x_t | x_{t-1}) g_t(x_t) dx_t} \log \left(\frac{g_t(x_t)}{h_t(x_t)} \right) dx_t \\ &= \int p^\circ(x_t | x_{t-1}) \log \left(\frac{g_t(x_t)}{h_t(x_t)} \right) dx_t, \end{aligned}$$

where the inequality follows from the logsum-inequality given in Lemma 5. $\square$

Corollary 2. If $c_t \leq \xi_t(x) \leq C_t$, then

$$\text{KL}(\mathbb{P}_T^*, \mathbb{P}_T^\circ) \leq \sum_{t=0}^T (C_t - c_t).$$

The condition in this corollary is admittedly very strong.

Corollary 3. We have

$$\text{KL}(\mathbb{P}_T^* \| \mathbb{P}_T^\circ) \leq \sum_{t=0}^T \left(\mathbb{E}^* \xi_t(X_t) - \sup_{x_{t-1}} \frac{p^*(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ [\min(\xi_t(X_t), 0)] \right), \quad (3)$$

where $p^*(x_{t-1})$ and $p^\circ(x_{t-1})$ denote the marginal densities of x_{t-1} under $\mathbb{P}_T^*$ and $\mathbb{P}_T^\circ$ respectively.

Proof. We have

$$\text{KL}(\mathbb{P}_T^* \| \mathbb{P}_T^\circ) \leq \sum_{t=0}^T \mathbb{E}^* \xi_t(x_t) - \sum_{t=0}^T \int \int p^\circ(x_t | x_{t-1}) \xi_t(x_t) dx_t p^*(x_{t-1}) dx_{t-1}.$$

The integral in the second term can be written as (divide and multiply by $p^\circ(x_{t-1})$ in the integral)

$$C := - \int \int \frac{p^*(x_{t-1})}{p^\circ(x_{t-1})} p^\circ(x_{t-1}, x_t) \xi_t(x_t) dx_t dx_{t-1}.$$

Let

$$A = \{x_t : \xi_t(x_t) \geq 0\}.$$

We have

$$C \leq \int_{x_{t-1}} \int_{x_t \in A^c} p^\circ(x_{t-1}, x_t) (-\xi_t(x_t)) dx_t \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} dx_{t-1}$$

as the integral will be negative when $x_t \in A$. Using Fubini, This leads to

$$\begin{aligned} C &\leq \sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \int_{x_t \in A^c} p^\circ(x_t) (-\xi_t(x_t)) dx_t \\ &= \sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ [-\xi_t(X_t) \mathbf{1}_{\{-\xi_t(X_t) > 0\}}] \\ &= -\sup_{x_{t-1}} \frac{p^\star(x_{t-1})}{p^\circ(x_{t-1})} \mathbb{E}^\circ [\min(\xi_t(X_t), 0)]. \end{aligned}$$

In [JHJ21] it seems that $A = \emptyset$ is assumed. In that case, the term $\mathbb{E}^\circ \min(\xi_t(X_t), 0)$ can be replaced by $\mathbb{E}^\circ \xi_t(X_t)$. □

In specific cases where $\mathbb{P}_T^\star$ is tractable, we may obtain sharper bounds.

Example 4. Gaussian case...

3.1. On the proof in appendix B2 in [JHJ21]. This is about upper bounding $p^\star(x_t)/p^\circ(x_t)$. They argue that it suffices to bound $p^\star(x_{0:T})/p^\circ(x_{0:T})$. **Why does this suffice?** By (1) this boils down to bounding each of the terms in $\Psi(x_{0:n})$. Their argument, translated to our notation, is that h_t is bounded and then $\int p(x_t | x_{t-1} h_t(x_t) dx_t$ is bounded. Then, it suffices to lower bound $\int p(x_t | x_{t-1} g_t(x_t) dx_t$. Their proof for this is specific to their setting.

3.2. On the proof in appendix B3 in [JHJ21]. This is about further bounding the bound in (3). In our notation, these bounds look like

$$\begin{aligned} \mathbb{E}^\star \log \xi_t(X_t) &\leq \sum_{k=t}^{T-1} c_{k+1}^\star \sqrt{\mathbb{E}^\star \varphi^\star(X_t)^2} \sqrt{\mathbb{E}^\star \Delta(X_t)} \\ -\mathbb{E}^\circ \log \xi_t(X_t) &\leq \sum_{k=t}^{T-1} c_{k+1} \sqrt{\mathbb{E}^\circ \varphi(X_t)^2} \sqrt{\mathbb{E}^\circ \Delta(X_t)}, \end{aligned}$$

where φ , $\varphi^\star$ and Δ are discrepancy measures of parameters in the true and approximate backward filters.

4. A TOY EXAMPLE: STOCHASTIC VOLATILITY MODEL

Consider a state-space model where the observations are given by the random sequence

$$R_i = \exp\left(\frac{X_i}{2}\right) U_i, \quad 1 \leq i \leq n, \quad (4)$$

and the latent states by

$$X_i = \omega + \psi X_{i-1} + \eta W_i, \quad 1 \leq i \leq n, \quad (5)$$

where $(U_i)_{i=1}^n, (W_i)_{i=1}^n$ are independent random sequences of i.i.d. standard normal variates and X has initial distribution $X_0 \sim \mathcal{N}\left(\frac{\omega}{1-\psi}, \frac{\eta^2}{1-\psi^2}\right)$. We impose the parameter constraints $\omega \in \mathbb{R}$, $-1 < \psi < 1$ and $\eta > 0$.

We consider the following transformed version of the observation equation (4)

$$V_i := \log(R_i^2) = X_i + U'_i, \quad 1 \leq i \leq n, \quad (6)$$

where $U'_i := \log(U_i^2)$, $i = 1, \dots, n$.

4.1. Backward Filtering. The proposed model is simple in the sense that the latent process is linear Gaussian. If the distribution of U'_i would be Gaussian, the backward information filter provides a recursion that allows exact computation of $\{h_i\}$. This is unfortunately not the case, though can obtain guiding functions $\{g_i\}$ by backward filtering under the assumption

$$V_i \mid X_i \sim \mathcal{N}(\mu + X_i, \sigma^2) \quad (7)$$

with $\mu := \mathbb{E}(U'_i) \approx -1.27$ and $\sigma^2 := \text{Var}(U'_i) = \pi^2/2$. In that case, the derivations are simple. It turns out to be useful to introduce the notation

$$U(c, F, H)(x) = \exp\left(c + Fx - \frac{1}{2}Hx^2\right). \quad (8)$$

It turns out that (7) implies any g_i can be expressed in this form for a triplet (c_i, F_i, H_i) .

- *Leaf Messages* From Theorem 6.1 (ii) in [VdMSS25], if v_n is observed at a leaf, then $g_n(x) = U(c_n, F_n, H_n)(x)$ with $c_n = \log(\varphi(v_n; \mu, \sigma^2))$, $F_n = (v_n - \mu)\sigma^{-2}$ and $H_n = \sigma^{-2}$.
- *Pullback Kernels* By statement (iii) of the same theorem, we have for the pullback $(\tilde{\kappa}_{n-1,n}, g_n)(x) = U(\bar{c}_n, \bar{F}_n, \bar{H}_n)(x)$ with

$$\begin{aligned} \bar{H}_n &= \psi^2 / C_n \\ \bar{F}_n &= \psi(F_n / H_n - \omega) C_n^{-1} \\ \bar{c}_n &= c_n - \log(\varphi^{\text{can}}(0; F_n, H_n)) + \log(\varphi(\omega; F_n / H_n, C_n)), \end{aligned}$$

where $C_n = \eta^2 + H_n^{-1}$.

Here

$$k(x, v) = \exp(v - x) f_{\chi^2(1)}(\exp(v - x)).$$

This is bounded since for $u > 0$, the density of $u f_{\chi^2(1)}(u)$ behaves like $u u^{-1/2} e^{-u/2}$ which is bounded.

APPENDIX A. SOME RESULTS

Lemma 5. Writing $a = \int a(x) dx$ and $b = \int b(x) dx$, the integral analogue of the log-sum inequality is

$$a \log\left(\frac{a}{b}\right) \leq \int a(x) \log\left(\frac{a(x)}{b(x)}\right) dx,$$

Ssee e.g. its use in [CS04], Lemma 7.4.

APPENDIX B. FEYNMAN-KAC MODELS

B.1. Classical Feynman-Kac models (on chains). In the classical setting, a Feynman-Kac model describes a sequence of random variables $(X_0, X_1, \dots, X_n)$ evolving according to:

- Markov transitions: $X_{k+1} \sim M_k(X_k, \cdot)$
- Potential functions: $G_k(x_{k-1}, x_k)$

The associated Feynman-Kac measure on paths is defined as:

$$\mathbb{Q}_n(dx_{0:n}) = \frac{1}{Z_n} \left[\prod_{k=0}^n G_k(x_{k-1}, x_k) \right] \mathbb{M}(dx_{0:n})$$

where $\mathbb{M}(dx_{0:n})$ is the law of the Markov chain, and Z_n is the normalizing constant.

Feynman-Kac models on directed trees. In a directed tree, the process evolves on a rooted tree T , where:

- Each node $t \in V(T)$ corresponds to a random variable X_t
- Each node t (except the root) has a parent $\text{pa}(t)$, and a transition kernel $M_t(x_{\text{pa}(t)}, \cdot)$
- Each node has a potential function $G_t(x_{\text{pa}(t)}, x_t)$

The root r has an initial distribution: $X_r \sim \mu_r$

Definition. The Feynman-Kac measure $\mathbb{Q}_T$ on the state space $\{X_t : t \in T\}$ is given by:

$$\mathbb{Q}_T(d\mathbf{x}) = \frac{1}{Z_T} \left[\prod_{t \in T} G_t(x_{\text{pa}(t)}, x_t) \right] \mathbb{M}_T(d\mathbf{x})$$

where $\mathbb{M}_T$ is the law of the Markov process over the tree:

$$\mathbb{M}_T(d\mathbf{x}) = \mu_r(dx_r) \prod_{t \neq r} M_t(x_{\text{pa}(t)}, dx_t)$$

and Z_T is the normalizing constant (partition function):

$$Z_T = \int \left[\prod_{t \in T} G_t(x_{\text{pa}(t)}, x_t) \right] \mathbb{M}_T(d\mathbf{x})$$

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